

Assignment:-1

1. What is a logic gate?

Ans: - A logic gate is an electronic circuit that performs a logical operation on one or more binary inputs and produces a single binary output. It follows Boolean algebra rules. Basic gates include AND, OR, and NOT. They are the building blocks of digital circuits. Used in processors, memory devices, and digital logic design.

2. What does an AND gate do?

Ans: - An AND gate is a logic gate that outputs true (1) only if all of its inputs are true (1); otherwise, it outputs false (0).

In terms of binary:

- If both inputs are 1 (true) → Output is 1 (true)
- If any one or both inputs are 0 (false) → Output is 0 (false)

Truth Table:

Input A	Input B	Output (A AND B)
0 (false)	0 (false)	0 (false)
0 (false)	1 (true)	0 (false)
1 (true)	0 (false)	0 (false)
1 (true)	1 (true)	1 (true)

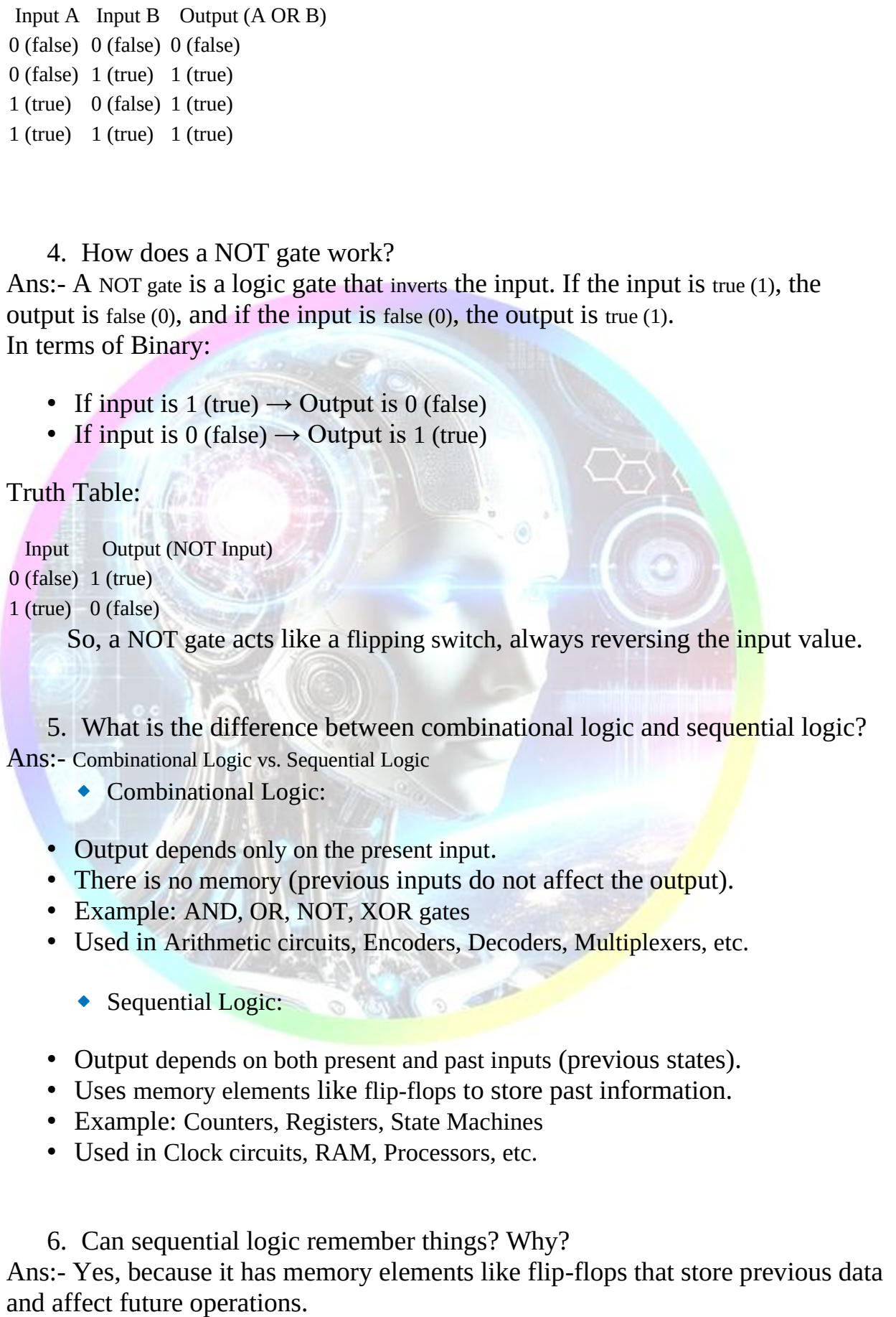
3. What is an OR gate used for?

Ans:- OR gate is a logic gate that outputs true (1) if at least one of its inputs is true (1). It only outputs false (0) when all inputs are false (0).

In terms of Binary:

- If both inputs are 0 (false) → Output is 0 (false)
- If at least one input is 1 (true) → Output is 1 (true)

Truth Table:



Input A	Input B	Output (A OR B)
0 (false)	0 (false)	0 (false)
0 (false)	1 (true)	1 (true)
1 (true)	0 (false)	1 (true)
1 (true)	1 (true)	1 (true)

4. How does a NOT gate work?

Ans:- A NOT gate is a logic gate that inverts the input. If the input is true (1), the output is false (0), and if the input is false (0), the output is true (1).

In terms of Binary:

- If input is 1 (true) → Output is 0 (false)
- If input is 0 (false) → Output is 1 (true)

Truth Table:

Input	Output (NOT Input)
0 (false)	1 (true)
1 (true)	0 (false)

So, a NOT gate acts like a flipping switch, always reversing the input value.

5. What is the difference between combinational logic and sequential logic?

Ans:- Combinational Logic vs. Sequential Logic

◆ Combinational Logic:

- Output depends only on the present input.
- There is no memory (previous inputs do not affect the output).
- Example: AND, OR, NOT, XOR gates
- Used in Arithmetic circuits, Encoders, Decoders, Multiplexers, etc.

◆ Sequential Logic:

- Output depends on both present and past inputs (previous states).
- Uses memory elements like flip-flops to store past information.
- Example: Counters, Registers, State Machines
- Used in Clock circuits, RAM, Processors, etc.

6. Can sequential logic remember things? Why?

Ans:- Yes, because it has memory elements like flip-flops that store previous data and affect future operations.

7. What is a flip-flop in a computer?

Ans:- A flip-flop is a small memory unit that stores one bit (0 or 1) and is used in sequential circuits.

8. What do flip-flops store in a computer?

Ans:- Flip-flops store binary data (0 or 1) and help in memory storage, registers, and data transfer inside a computer.

9. Can you give an example of a combinational logic circuit?

Ans:- Yes, an adder circuit is a combinational circuit used for arithmetic operations like addition in digital systems.

10. What happens in a sequential logic circuit?

Ans:- The output depends on both current and past inputs, making it useful for memory storage and counters.

11. What is a register in a computer?

Ans:- A register is a fast storage unit inside the CPU that holds data temporarily for processing and execution.

12. Why does the CPU need registers?

Ans:- Registers store temporary data, improving processing speed by reducing memory access delays.

13. What is the Program Counter (PC) register?

Ans:- The PC register holds the address of the next instruction to be executed by the CPU.

14. What does the Accumulator register do?

Ans:- The accumulator holds intermediate results of arithmetic and logical operations performed by the CPU.

15. What is the Instruction Register (IR) in a computer?

Ans:- The IR stores the current instruction being executed by the CPU for processing.

16. How does the CPU know which instruction to execute next?

Ans:- The CPU follows the Program Counter (PC), which keeps track of the next instruction's memory address.

17. Why are registers faster than memory?

Ans:- Registers are built using high-speed storage, reducing access time compared to RAM and other memory types.

18.Can you name a type of register used in a computer?

Ans:- Yes, the Instruction Register (IR) stores the instruction currently being processed by the CPU.

19.What does the Control Unit use registers for?

Ans:- The Control Unit uses registers to fetch, decode, and execute instructions efficiently.

20.What happens when a register gets full?

Ans:- Data moves to RAM or cache, freeing up space for new instructions.

21.What does the control unit do in a computer?

Ans:- It manages data flow, executes instructions, and controls hardware operations.

22.What is the purpose of the clock in a computer?

Ans:- The clock in a computer controls the timing of all operations, making sure tasks happen in a synchronized and orderly way.

23.What is timing in computer systems?

Ans:- Timing in computer systems refers to when and how fast operations are performed, ensuring everything runs in a coordinated manner.

24.Why do we need a timing system for the CPU?

Ans:- It ensures smooth execution of instructions without conflicts or errors.

25.How does the control unit help in fetching instructions?

Ans:- It retrieves instructions from memory and sends them for execution.

26.What is an instruction cycle in a computer?

Ans:- The cycle includes fetching, decoding, executing, and storing results of an instruction.

27.What happens during the fetch part of the instruction cycle?

Ans:- The CPU retrieves the instruction from memory using the Program Counter (Next Instruction Tracker)

28.What happens during the decode part of the instruction cycle?

Ans:- The instruction is interpreted to determine what operation needs to be performed.

29.What happens during the execute part of the instruction cycle?

Ans:- During the **execute** part, the CPU **performs the instruction's task**, like calculations, data movement, or decision-making, based on the instruction code.

30. Why does the CPU repeat the instruction cycle?

Ans:- The CPU repeats the instruction cycle to continuously fetch, decode, and execute instructions, ensuring the computer runs programs smoothly.

31. What is a memory reference instruction in a computer?

Ans:- A memory reference instruction is a command that accesses data from memory (read/write) during execution, like LOAD, STORE, or MOVE.

32. Why do memory reference instructions help in reading and writing data?

Ans:- Memory reference instructions help in reading and writing data by allowing the CPU to access, store, or modify information in memory during execution.

33. How does the CPU find data in memory using a memory reference instruction?

Ans:- The CPU finds data in memory using a memory reference instruction by accessing the memory address where the required data is stored.

34. What is the difference between a memory reference instruction and a regular instruction?

Ans:- A memory reference instruction accesses data from memory, while a regular instruction may perform operations like arithmetic or logic without needing memory access.

35. How does the computer access data in memory?

Ans:- A computer accesses data in memory using the memory address, which helps locate and retrieve stored information when needed.

36. What is an I/O interrupt in a computer?

Ans:- An I/O interrupt is a signal sent by an external device (keyboard, printer) to the CPU, requesting attention. It allows the CPU to pause its current task and handle the device request. This prevents continuous CPU monitoring of input/output devices. The interrupt helps in efficient multitasking and system responsiveness. It is crucial in operating systems for smooth execution.

37. Why does the CPU need to handle I/O interrupts?

Ans:- I/O interrupts help the CPU manage multiple tasks without waiting for devices. Without them, the CPU would waste time checking for input/output requests. Handling interrupts improves system performance and multitasking.

ability. It ensures the CPU quickly responds to external events. This process helps in efficient resource utilization.

38. What happens when a keyboard sends an interrupt to the CPU?

Ans:- When a key is pressed, the keyboard sends an interrupt signal. The CPU pauses its current execution and processes the keystroke. It then stores the input in a Temporary Storage for further processing. After handling the interrupt, the CPU resumes its previous task. This process ensures fast and smooth user interaction.

39. How does the CPU know when to handle an interrupt?

Ans:- The CPU constantly checks the Interrupt Request (IRQ) line. When a device sends an interrupt signal, the CPU detects it. The CPU stops its current task and switches to the interrupt handler. It processes the request and completes the necessary action. After that, the CPU returns to the original task.

40. What happens after the CPU finishes handling an interrupt?

Ans:- The CPU updates necessary registers and clears the interrupt flag. It restores the previous execution state from the stack. Then, it resumes the task it was performing before the interrupt. If multiple interrupts occur, they are handled based on priority. This ensures smooth and efficient processing.

